

ALTA, Norway

WMO: 010490

Lat: 69.976N

Lon: 23.372E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.0	-18.8	-24.8	0.4	-20.7	-22.6	0.5	-18.4	13.3	-0.6	11.3	-1.0	3.2	210	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	22.6	15.4	20.4	14.7	18.5	13.8	16.3	21.0	15.2	19.5	14.2	18.0	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.0	10.0	18.8	13.0	9.3	17.5	12.0	8.8	16.2	45.7	21.2	42.5	19.5	39.7	18.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.2	7.9	DB	-23.1	26.2	3.2	2.5	-25.4	28.0	-27.2	29.5	-29.0	30.9	-31.3	32.7
			WB	-23.4	18.0	3.1	1.6	-25.6	19.1	-27.4	20.1	-29.1	21.0	-31.4	22.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-7.2	-7.4	-4.1	0.3	5.5	10.2	13.8	12.4	8.4	2.6	-2.8	-4.8
	DBStd	8.57	6.04	6.21	4.95	3.80	3.51	3.15	3.25	3.01	3.32	3.98	4.62	5.47
	HDD10.0	3078	532	487	437	290	146	33	3	8	68	230	384	459
	HDD18.3	5854	791	720	696	540	398	243	146	186	297	487	634	717
	CDD10.0	274	0	0	0	0	7	41	121	83	22	1	0	0
	CDD18.3	8	0	0	0	0	0	1	6	2	0	0	0	0
	CDH23.3	48	0	0	0	0	0	3	37	9	0	0	0	0
	CDH26.7	8	0	0	0	0	0	0	7	1	0	0	0	0
(14) Wind	WSAvg	3.6	3.9	3.9	3.8	3.4	3.6	3.8	3.6	3.2	3.1	3.3	3.5	3.9
(15) Precipitation	PrecAvg	436	36	26	30	24	28	41	53	50	38	44	31	35
	PrecMax	590	72	50	52	49	61	90	155	98	74	83	73	88
	PrecMin	292	7	7	6	6	4	7	7	19	13	7	1	10
	PrecStd	69	18	11	12	11	16	20	32	20	15	20	17	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.3	7.2	11.2	18.8	22.8	27.3	24.5	18.8	13.4	8.0	8.0
		MCWB	3.4	3.4	3.9	6.2	11.4	14.2	18.3	16.5	13.8	9.8	5.4	4.8
	2%	DB	4.2	4.9	5.3	8.7	15.2	19.6	23.7	21.5	16.8	11.1	6.1	5.6
		MCWB	2.0	2.4	2.4	5.1	9.5	13.1	16.1	15.3	12.3	8.1	4.0	3.0
	5%	DB	2.4	2.9	3.8	7.0	12.8	17.2	21.3	19.3	15.0	9.2	4.5	3.8
		MCWB	0.5	0.9	1.3	4.0	8.0	12.0	15.2	14.4	11.6	6.6	2.6	1.6
	10%	DB	0.8	1.0	2.3	5.5	10.7	15.2	19.2	17.4	13.4	7.9	3.2	2.4
		MCWB	-0.8	-0.6	0.1	2.8	6.7	10.9	14.7	13.6	10.4	5.8	1.5	0.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.8	4.1	6.7	12.0	15.4	18.8	17.6	14.2	10.1	5.6	5.0
		MCDB	5.9	6.0	6.9	10.3	17.7	19.9	25.4	22.2	18.0	13.2	7.5	7.6
	2%	WB	2.0	2.4	2.7	5.2	9.9	13.8	16.9	16.0	12.8	8.3	4.0	3.0
		MCDB	4.0	4.6	5.1	8.2	14.8	18.9	22.3	20.3	15.8	10.8	5.9	5.2
	5%	WB	0.6	0.9	1.4	4.1	8.3	12.5	15.7	14.8	11.7	6.9	2.7	1.7
		MCDB	2.2	2.7	3.5	6.6	12.2	16.7	20.3	18.8	14.5	8.8	4.3	3.5
	10%	WB	-0.8	-0.6	0.3	3.0	7.0	11.3	14.8	13.8	10.6	5.9	1.6	0.6
		MCDB	0.6	0.9	2.2	5.3	10.2	14.7	18.9	17.1	13.1	7.7	3.1	2.2
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.0	6.6	6.1	5.8	5.5	5.9	6.0	5.5	4.6	5.1	5.7
		MCDBR	6.6	6.5	5.9	6.8	9.2	8.9	9.9	9.8	7.4	5.5	5.3	6.1
		MCWBR	5.5	5.3	4.5	4.5	5.2	4.7	4.7	4.9	4.4	3.8	4.4	4.9
	5% WB	MCDBR	6.5	6.5	5.8	6.3	8.7	8.2	8.9	8.6	6.7	5.2	5.1	6.2
		MCWBR	5.6	5.5	4.7	4.3	5.0	4.8	4.6	4.7	4.2	3.8	4.5	5.2
(40) Clear-Sky Solar Irradiance	taub	N/A	0.248	0.300	0.323	0.339	0.341	0.349	0.334	0.306	0.258	N/A	N/A	
	taud	N/A	2.044	2.115	2.178	2.303	2.464	2.478	2.503	2.533	2.191	N/A	N/A	
	Ebn at Noon	0	459	686	802	839	854	831	802	725	452	0	0	
	Edh at Noon	0	41	79	102	103	91	87	75	55	34	0	0	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.30	1.50	3.40	4.76	4.86	4.44	3.07	1.60	0.47	0.00	0.00	
	RadStd	0.00	0.03	0.11	0.24	0.41	0.43	0.45	0.26	0.15	0.05	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page